

DISTORTED BY GRAVITY

Most of the bright objects in this image are galaxies in the Abell 2218 cluster. The arcs are much more remote galaxies, their images distorted by Abell 2218's gravity.

shifted into the infrared part of the spectrum. At the time, it was the most distant galaxy known, at 13 billion light-years from Earth. It must have formed shortly after the first stars, in the aftermath of the Big Bang.

Gravitational lensing can also reveal hidden properties of Abell 2218 itself. Because the strength of lensing depends on the cluster's density, it offers a measure of the distribution of all matter in the cluster—including the dark matter. Abell 2218 is one of the few galaxy clusters in which the pattern of visible matter (galaxies and X-ray-emitting gas) and the calculated distribution of dark matter do not match, suggesting the cluster is not as uniform as it appears in visible light.

Astronomers have now begun to use Abell 2218 to probe the origins of the universe. A phenomenon called the Sunyaev-Zel'dovich effect (see caption, opposite) creates holes and ripples in the cosmic microwave background radiation shining through the cluster. This happens because gas around Abell 2218's core scatters photons of microwave radiation, just as Earth's atmosphere scatters light. The strength of these ripples can be used to estimate the true diameter of the cluster's core, and therefore its distance from Earth, independently of its red shift. The red shift and distance can then be used together to find the expansion rate of the universe (see p.44).

EXPLORING SPACE**MAPPING THE MISSING MASS**